RAJALAKSHMI ENGINEERING COLLEGE

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S2

Started on	Tuesday, 25 August 2026, 9:27 AM
State	Finished
Completed on	Tuesday, 25 August 2026, 9:28 AM
Time taken	1 min 8 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using the counter method.

```
void func(int n)
{
    if(n==1)
    {
      printf("*");
    }
    else
    {
     for(int i=1; i<=n; i++)
     {
       for(int j=1; j<=n; j++)
       {
          printf("*");
          printf("*");
          break;
       }
     }
   }
 }
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>

int main() {
    int n;
    scanf("%d", &n);

    int count = 0;

    if (n == 1) {
        printf("%d", count);
    } else {
        count++;                // i = 1 (outer for initialization)

        int i = 1;
        while (1) {
            count++;            // i <= n
            if (i > n) break;

            count++;            // j = 1 (inner for initialization)
            int j = 1;

            while (1) {
                count++;        // j <= n
                if (j > n) break;

                count++;        // break
                break;
            }
```

```
29
30              i++;
31              count++;              // i++
32          }
33
34          printf("%d", count);
35      }
36
37      return 0;
38  }
```

	Input	Expected	Got	
✔	2	12	12	✔
✔	1000	5002	5002	✔
✔	143	717	717	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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